The Second Moment of ω under Twin-Centre Conditioning on the $6N$ Skeleton: A Near-Poisson Tilt with Conserved Dispersion

Ruqing Chen
GUT Geoservice Inc., Montreal, Quebec, Canada
ruqing@hotmail.com

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Abstract

Every earlier part of this programme worked with *first* moments: the mean enrichment of twin centres as a function of $\omega_{>3}(N)$, the number of distinct prime factors of the centre exceeding 3. The *second* moment—the Erdős–Kac-type variance, and how it responds to conditioning on a centre being a twin centre—was left open. We close it empirically. Sieving all centres up to $6N = 6 \times 10^8$ (2,166,300 twin centres), we measure the full distribution of $\omega_{>3}(N)$ over all centres and over twin centres. Twin-conditioning shifts the mean by a saturating amount ($+0.171 \to +0.215$ as $6N$ runs $1.5\times10^6 \to 6\times10^8$, decelerating toward a finite, small-prime-dominated limit) and inflates the variance by a stable factor 1.104 ± 0.005, while the dispersion ratio Var/mean is *conserved*: its twin-to-all ratio is 1.011 ± 0.004 across two and a half orders of magnitude. We interpret this as a *near-Poisson tilt*: writing ω as a sum of nearly independent Bernoulli indicators $\mathbf{1}[q \mid N]$, the twin weight $\prod_{q|N} q/(q-2)$ nudges each inclusion probability up by $\approx 2/(q(q-2))$, a convergent small-prime-dominated series that shifts the mean to a finite limit and rescales mean and variance nearly in step. The Erdős–Kac dispersion thus survives conditioning. We are explicit: this is new *to the programme*, not new mathematics—it confirms, and does not supersede, classical Erdős–Kac. We also record a clarification that ties off a separate thread: the "annihilation lattice" of Part XXI is exactly the classical Hardy–Littlewood admissibility condition.

1 Introduction

The arithmetic-geodynamics programme has repeatedly used the stratification of centres N by $\omega_{>3}(N)$, the count of distinct prime factors > 3: twin centres are enriched at high ω (the "wind tunnel" of Volume I and the kinetics of Part XXIII) [3, 5]. Every such statement was a statement about the mean. The distribution itself—in particular its variance, the object of the Erdős–Kac theorem [1]—was never measured here, and the variance was flagged as open throughout.

The classical Erdős–Kac theorem says $\omega(n)$ is asymptotically normal with mean and variance both $\sim \ln\ln n$. The programme-specific question is what happens to this distribution when we *condition* on N being a twin centre. The first-moment results guarantee the mean moves up; they say nothing about the spread. This paper measures both, at scale.

2 Method

For a centre N let $\omega_{>3}(N)$ be the number of distinct primes $p > 3$ dividing N (i.e. $\omega(N)$ minus the indicators of $2 \mid N$ and $3 \mid N$). A centre is a *twin centre* if $6N-1$ and $6N+1$ are both prime. We sieve all centres $N \le 10^8$ (so $6N \le 6 \times 10^8$), computing $\omega_{>3}(N)$ by a distinct-prime-factor sieve and twin-centre membership from a prime sieve to 6×10^8, and we tabulate the empirical distribution of $\omega_{>3}$ over all centres and over the 2,166,300 twin centres, at a ladder of cutoffs.

3 Results

The two distributions. Figure 1 shows the distribution of $\omega_{>3}(N)$ over all centres versus over twin centres at $6N = 6 \times 10^8$. The twin distribution is shifted right and modestly widened; the per-ω enrichment $P(\omega \mid \text{twin})/P(\omega \mid \text{all})$ rises monotonically across $\omega = 1, \dots, 5$, the distributional face of the multiplicative wind-tunnel enrichment.

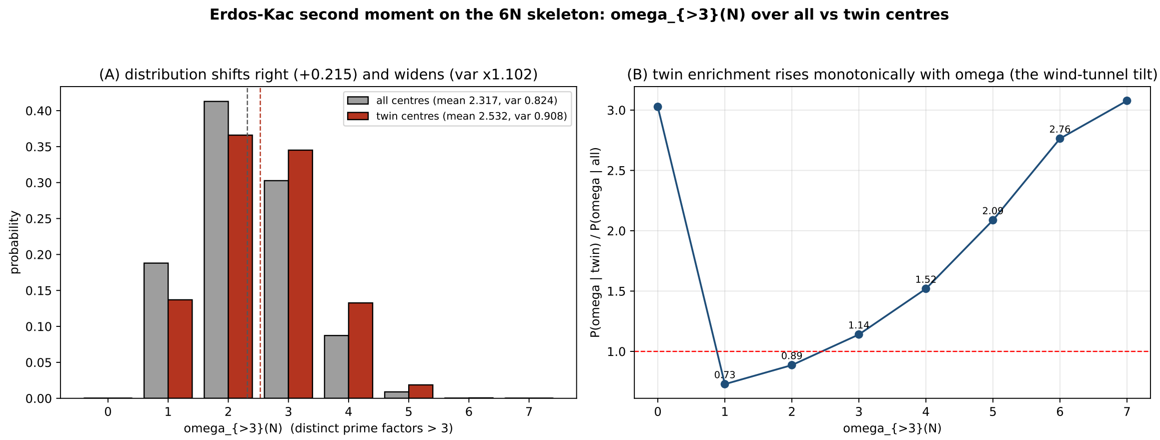

Figure 1: (A) Distribution of $\omega_{>3}(N)$ over all centres vs. twin centres at $6N = 6 \times 10^8$: a rightward mean shift and a modest widening. (B) The per-ω enrichment ratio rises monotonically over $\omega = 1$–5 (the wind-tunnel tilt); $\omega = 0, 6$ are low-count edge bins.

Moments across scale. Table 1 collects the first two moments at six cutoffs. Two trends are clean and stable (Fig. 2). First, the *mean shift* $\langle\omega\rangle_{\text{twin}} - \langle\omega\rangle_{\text{all}}$ grows but decelerates sharply, approaching a finite limit ≈ 0.21–0.22; it does not track $\ln\ln(6N)$. Second, the *dispersion ratio* Var/mean rises with scale for both populations (finite-size approach to the Erdős–Kac value), but the twin-to-all ratio of Var/mean stays at 1.011 ± 0.004, and the variance ratio at 1.104 ± 0.005, over the full range.

$6N_{\max}$	$\ln\ln(6N_{\max})$	mean shift	Var ratio (twin/all)	$(\text{V/m})_{\text{all}}$	$(\text{V/m})_{\text{twin}}$
1.5×10^6	2.655	+0.1710	1.097	0.2896	0.2916
6×10^6	2.748	+0.1845	1.109	0.3066	0.3117
3×10^7	2.846	+0.1995	1.111	0.3252	0.3303
1×10^8	2.913	+0.2085	1.105	0.3380	0.3414
3×10^8	2.971	+0.2120	1.102	0.3489	0.3518
6×10^8	3.006	+0.2149	1.102	0.3554	0.3585

Table 1: First two moments of $\omega_{>3}(N)$ over all vs. twin centres, at six cutoffs. The mean shift saturates; the variance ratio and the dispersion ratio are scale-stable.

4 Interpretation: a near-Poisson tilt

Write $\omega_{>3}(N) = \sum_{q>3} \mathbf{1}[q \mid N]$, a sum of nearly independent Bernoulli indicators with $\Pr[q \mid N] = 1/q$. The two-centre enrichment established earlier weights a centre by $w(N) = \prod_{q|N,\,q>3} q/(q-2)$ [5], so conditioning on a twin centre nudges each inclusion probability from $1/q$ toward $\sim 1/(q-2)$.

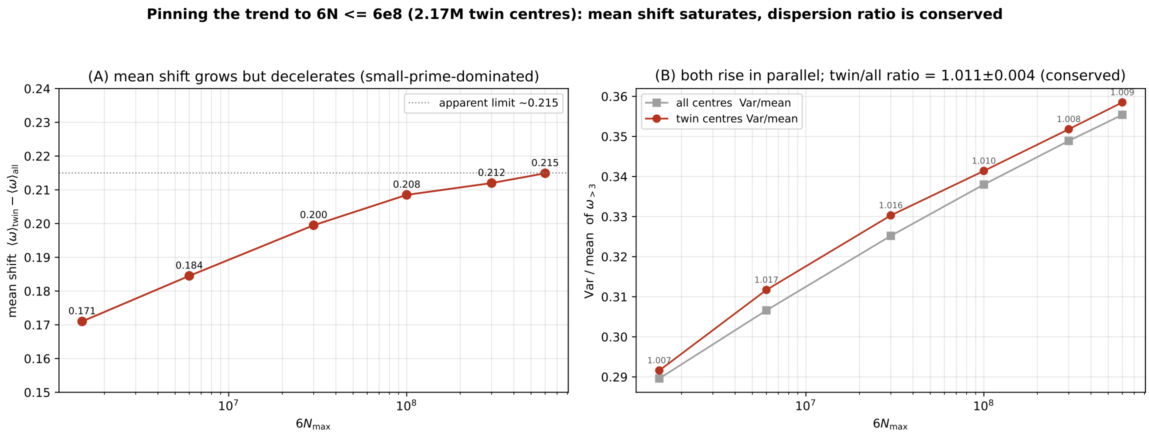

Figure 2: (A) The mean shift grows but decelerates toward an apparent limit ≈ 0.215 (small-prime-dominated). (B) Var/mean rises in parallel for both populations; their ratio is conserved at 1.011 ± 0.004 across $6N = 1.5 \times 10^6$ to 6×10^8.

Mean shift is a convergent small-prime sum. To leading order the shift in the mean is

$$\langle\omega\rangle_{\rm twin} - \langle\omega\rangle_{\rm all} \;\approx\; \sum_{q>3} \left(\tfrac{1}{q-2} - \tfrac{1}{q}\right) = \sum_{q>3} \frac{2}{q(q-2)}, \tag{1}$$

a convergent series dominated by the smallest primes ($q = 5$ alone contributes $\frac{2}{15} = 0.133$; $q = 7$ adds 0.057). The leading-order value ≈ 0.26 overestimates the measured shift because it ignores the normalisation by $\mathbb{E}[w]$ and inter-prime correlations; the measured shift approaches a limit of this order *from below*, which is exactly the observed saturation: the shift is set by a handful of small primes and cannot grow like $\ln\ln$.

Dispersion is conserved. For a sum of Bernoulli(p_q) indicators, mean $= \sum p_q$ and Var $= \sum p_q(1-p_q)$, so Var/mean $= 1 - \left(\sum p_q^2\right)/\left(\sum p_q\right)$. The twin tilt raises the p_q only slightly, and only for $q > 3$, so this ratio moves by about a percent—precisely the measured $(\mathrm{V/m})_{\rm twin}/(\mathrm{V/m})_{\rm all} = 1.011 \pm 0.004$. The tilt rescales mean and variance nearly in step: it is, to good approximation, a *near-Poisson tilt*, and the Erdős–Kac dispersion survives conditioning intact.

5 A clarification: the annihilation lattice is classical admissibility

A separate thread of Volume II asked for the "general k-body tiling spectrum." We record here, to prevent a misreading, that the tiling/annihilation criterion of Part XXI [4]—a configuration of constellations has zero joint density iff their shifted dead-residue sets cover $\mathbb{Z}/q$ for some prime q—is *identical* to the classical Hardy–Littlewood admissibility condition [2]: a set of offsets is inadmissible iff it occupies every residue modulo some prime. For instance $\{0, 2, 4\}$ covers $\mathbb{Z}/3$ (inadmissible = "annihilates"), while $\{0, 2, 6\}$ does not (admissible). The annihilation lattice is thus the complement of the admissible set, and it connects to Erdős covering systems; mapping it on the skeleton recovers known combinatorics in new coordinates, and we make no claim of a new criterion.

6 Conclusion

Measuring the second moment closes a question the programme had left open since Volume I. Conditioning on a twin centre acts as a near-Poisson tilt of the $\omega_{>3}$ distribution: it shifts the mean by a saturating, small-prime-dominated amount ($\to\approx 0.21$–0.22), inflates the variance by a stable factor 1.104, and conserves the dispersion ratio Var/mean to about a percent across two and a half orders of magnitude. The Erdős–Kac spread is preserved under conditioning, merely rescaled. We state the scope plainly: this is a new measurement *for this programme*, not a new theorem; it confirms classical Erdős–Kac rather than extending it, and it makes no claim about the infinitude of twins or any constellation. Together with the admissibility clarification, it ties off the two open threads of Volume II.

Data and code availability. The sieve experiment (to $6N = 6 \times 10^8$), the figures, and a verifier (small-scale moments and the admissibility=covering equivalence) are openly available at `https://github.com/Ruqing1963/6N-omega-variance` [7].

References

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